$\beta_2 = 0$，传输方程：

$$\frac{\partial U}{\partial z} = \frac{i e^{-\alpha z}}{L_{NL}} |U|^2 U$$

$L_{NL} \equiv (\gamma P_0)^{-1}$

α：损耗。

L_{NL}：非线性长度，γ：非线性参数，P_0：峰值功率。

$U = V \exp(i\phi_{NL})$，代入。

$$\frac{\partial V}{\partial z} e^{i\phi_{NL}} + iV e^{i\phi_{NL}} \frac{\partial \phi_{NL}}{\partial z} = \frac{i e^{-\alpha z}}{L_{NL}} V^2 e^{i\phi_{NL}}$$

$$\Leftrightarrow \quad \frac{\partial V}{\partial z} e^{i\phi_{NL}} + V \frac{\partial \phi_{NL}}{\partial z} e^{i(\phi_{NL}+\frac{\pi}{2})} = \frac{V^2}{L_{NL}} e^{-\alpha z} e^{i(\phi_{NL}+\frac{\pi}{2})}.$$

$$\begin{cases} \frac{\partial V}{\partial z} = 0. & \phi_{NL} \text{ 方向上}, \\ \frac{\partial \phi_{NL}}{\partial z} = \frac{V^2}{L_{NL}} e^{-\alpha z}. & \perp \phi_{NL} \text{ 方向上}, \end{cases}$$

- 振幅 V 不随 z 变化。
- 非线性相位 $\phi_{NL} \quad \alpha (1 - e^{-\alpha z})$。

通解：

$$\begin{cases} U(L,T) = U(0,T) \exp[i\phi_{NL}(L,T)]. \\ \phi_{NL}(L,T) = |U(0,T)|^2 \frac{L_{eff}}{L_{NL}} \\ L_{eff} = \frac{1 - e^{-\alpha L}}{\alpha} \quad \text{光纤有效长度}. \end{cases}$$

Note:

1. 自相位调制产生与 T 相关相移. 不改变脉冲形状.
2. $L\uparrow$ $\phi_{NL}\uparrow$. L_{eff} 考虑了光纤的损耗, 略小于 L.
3. $\alpha=0$, $L_{eff}=L$, 最大非线性相移 ϕ_{max} 出现在脉冲中心. ($T=0$).

$$\phi_{max}=\frac{L_{eff}}{L_{NL}}=\gamma P_0 L_{eff}.$$

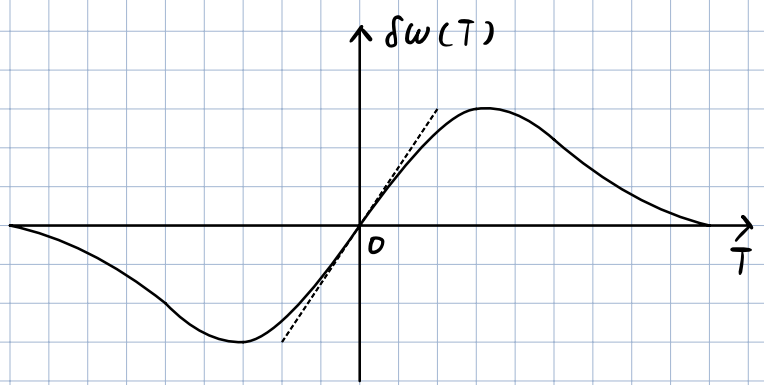
瞬时变化的相位 $\longrightarrow$ 瞬时光频率.

距中心频率 ω_0 差值

$$\delta\omega(T)=-\frac{\partial\phi_{NL}}{\partial T}=-\left(\frac{L_{eff}}{L_{NL}}\right)\frac{\partial}{\partial T}|U(0,T)|^2$$

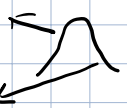
频率啁啾.

随脉冲传输, 新的频率分量不断产生.



$\delta\omega$：

1. $\delta\omega$ $\begin{cases} <0 : \text{红移，前沿} \\ >0 : \text{蓝移，后沿} \end{cases}$



2. 脉冲中央区域，啁啾线性且 >0.

3. 脉冲若有陡阴前后沿，chirp 显著↑.

4. 不同高斯脉冲，超高斯脉冲 chirp 仅发生于脉冲沿附近且非线性.

脉冲频谱变化？

无 chirp 情形.

$$\delta\omega(T) = \frac{2m}{T_0}\frac{L_{eff}}{L_{NL}}\left(\frac{T}{T_0}\right)^{2m-1}\exp\left[-\left(\frac{T}{T_0}\right)^{2m}\right].$$

$\delta\omega$ 最大值？

$$\frac{d\,\delta\omega}{dT} = 0 \quad \Rightarrow \quad \left[(2m-1)T^{2m-2} - T^{2m-1}\cdot\frac{2m}{T_0^{2m}}\cdot T^{2m-1}\right] = 0$$

$$\Leftrightarrow \qquad T^{4m}\cdot\frac{2m}{T_0^{2m}} - T^{2m}\cdot(2m-1) = 0$$

$$\Leftrightarrow \qquad T/T_0 = \left(1-\frac{1}{2m}\right)^{\frac{1}{2m}}$$

$$\delta\omega_{max} = \frac{2m}{T_0}\frac{L_{eff}}{L_{NL}}\left(1-\frac{1}{2m}\right)^{1-\frac{1}{2m}}\exp\left(\frac{1}{2m}-1\right).$$

$$= \frac{m f(m)}{T_0}\phi_{max}. \qquad \phi_{max} = \frac{L_{eff}}{L_{NL}}$$

$$f(m) = 2\left(1-\frac{1}{2m}\right)^{1-1/2m}\exp\left(\frac{1}{2m}-1\right).$$

For 无 chirp Gaussian pulse: $m=1$, 且 $\Delta\omega_0 = \frac{1}{T_0}$

$$\delta\omega_{max} = 0.86\,\Delta\omega_0\,\phi_{max}.$$